Catch and biology timing within a time step

A simulation to inform opal's design, using the IO yellowfin model

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1 Why this matters

Fitted to the same IO yellowfin data with the same growth, opal and SS3 differ in stock size and depletion. Two of the structural differences between them concern *when within a time step* things happen:

a. **Catch.** opal removes each quarter's catch instantaneously at the start of the quarter and then applies natural mortality. SS3 (`F_Method = 3`) uses the Baranov catch equation: fishing and natural mortality act together through the quarter.
b. **Biology.** opal evaluates everything at the start of the quarter: the length-at-age that maps its length-based selectivity onto ages and predicts the length compositions, and the abundance the index samples. SS3 uses the middle of the season for the catch's length-at-age and for the index.

This document isolates both with a small, self-contained model. It asks how much each convention changes the population an assessment implies for the same catch, and how much it biases B0 and depletion when the estimation model's convention differs from the truth. It also tests a third catch option, **Pope's mid-step approximation** (natural mortality for half the step, the catch removed instantaneously, then the other half), which is as cheap and explicit as opal's current method.

2 The model

A quarterly, single-region, age-structured model (ages 1–28 quarters, plus group), written in R and RTMB in `R/timing-model.R`. Its inputs come from the IO yellowfin conversion (`code/rtmb/opal-export`):

- **Biology:** M, maturity × fecundity and SS3′s start-of-quarter length-at-age (CV 0.1). Mid-quarter length is half-way to the next age.
- **Fisheries:** the 16 SS3 fleets are pooled by gear into three, with their observed catches in numbers summed. Selectivity is length-based (opal's double-normal curve) with C2′s fitted parameters for one representative fleet each:

fishery	SS3 fleets	selectivity from C2 fleet	total catch (millions)	estimated selectivity columns
purse_seine	11, 12, 13	11	408.9	1, 2, 3, 4
long-line	4, 5, 6, 7	4	94.2	1, 3
other	1, 2, 3, 8, 9, 10, 14, 15, 16	8	525.5	1, 2, 3, 4

- **Recruitment:** Beverton–Holt, steepness 0.8, σ_R 0.6, recruits at age 1 from SSB at the start of the next quarter (opal's timing), unfished start, B0 = C2′s estimate, 4.10 million t.

Each run picks one **catch method** and one **biology timing**:

Catch method	Removal within the quarter	Survival over the quarter
`start` (opal)	$u = C / \Sigma\, s\, N$ at the start	$(1 - H)\, e^{-M}$
`pope_mid`	$u = C / \Sigma\, s\, N\, e^{-M/2}$ at mid-step	$e^{-M/2}\,(1 - H)\, e^{-M/2}$
`baranov` (SS3)	F solved from $C = \Sigma\, F\, s\, N\, (1 - e^{-Z}) / Z$	e^{-Z}

Biology timing	Age–length key (selectivity-at-age, compositions)	Index abundance
`start`	start-of-quarter length	N at the start
`mid`	mid-quarter length	N × √(survival), half the quarter's mortality

H is the total harvest fraction at age ($\Sigma_f u_f s_{fa}$, smoothly capped at 0.95) and $Z = M + \Sigma_f F_f s_{fa}$. Baranov F is solved by fixed-point iteration from Pope's value (30 iterations in the operating model, 8 in the estimation models; SS3′s hybrid method uses a similar handful).

3 Part 1: the same catch, different conventions

Every combination is projected with the same parameters and the same catch, scaled by 0.25–1.25. Nothing is estimated, so the differences are purely structural.

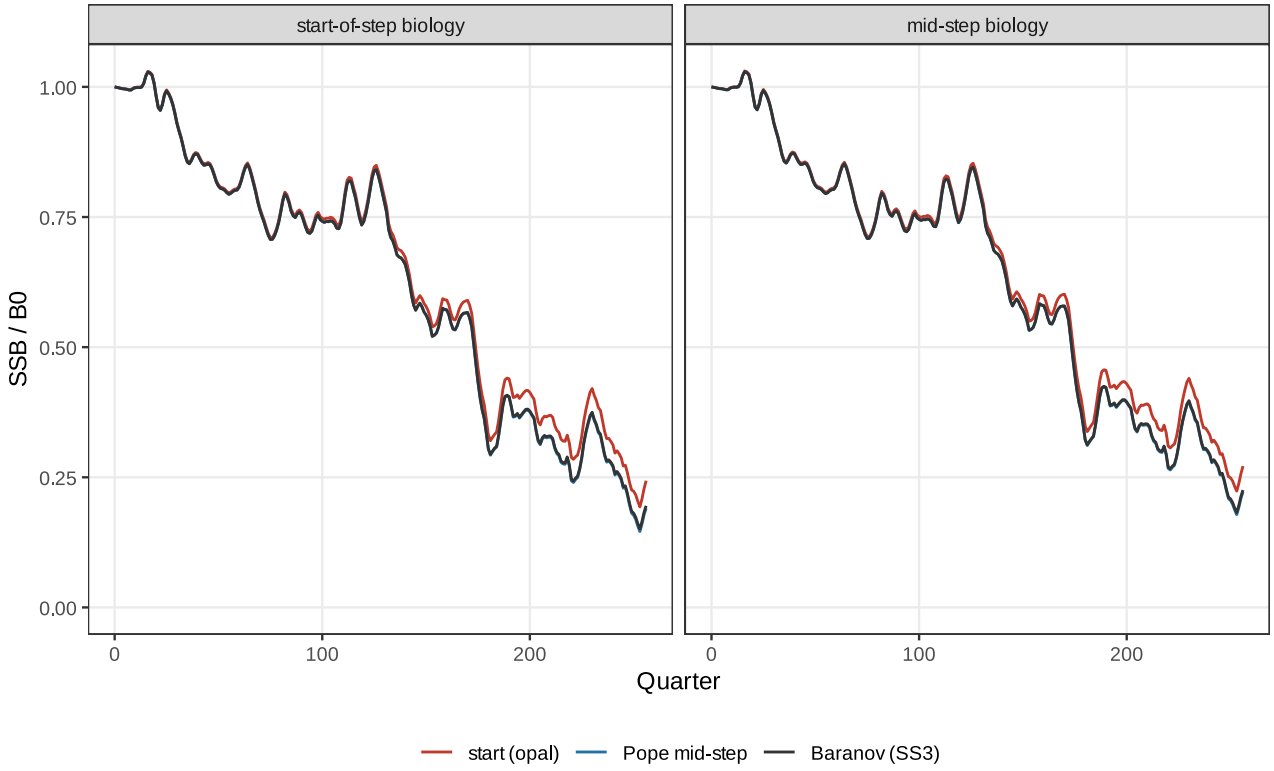


Figure 1: Static depletion SSB/B0 at the observed catch, for each catch method and biology timing, with identical parameters and catches.

Catch ×	Biology	start (opal)	Pope mid-step	Baranov (SS3)	start vs Baranov	Pope vs Baranov
0.25	start	0.739	0.726	0.726	2%	−0.0%
0.25	mid	0.746	0.733	0.734	2%	−0.0%
0.50	start	0.572	0.546	0.547	5%	−0.1%
0.50	mid	0.585	0.561	0.562	4%	−0.1%
0.75	start	0.408	0.371	0.373	10%	−0.5%
0.75	mid	0.429	0.393	0.394	9%	−0.4%
1.00	start	0.243	0.191	0.195	25%	−2.4%
1.00	mid	0.272	0.222	0.225	20%	−1.7%
1.25	start	0.042	0.000	0.000	collapse	collapse
1.25	mid	0.092	0.002	0.010	collapse	collapse

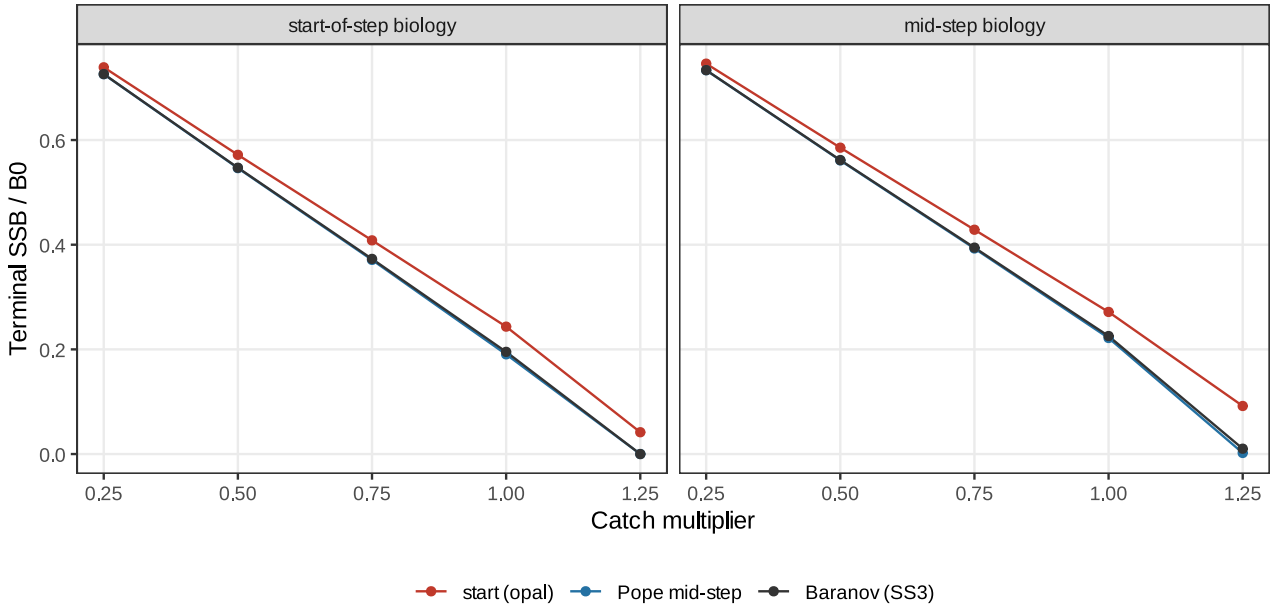


Figure 2: Terminal depletion against the catch multiplier. Points above 1 are close to collapse.

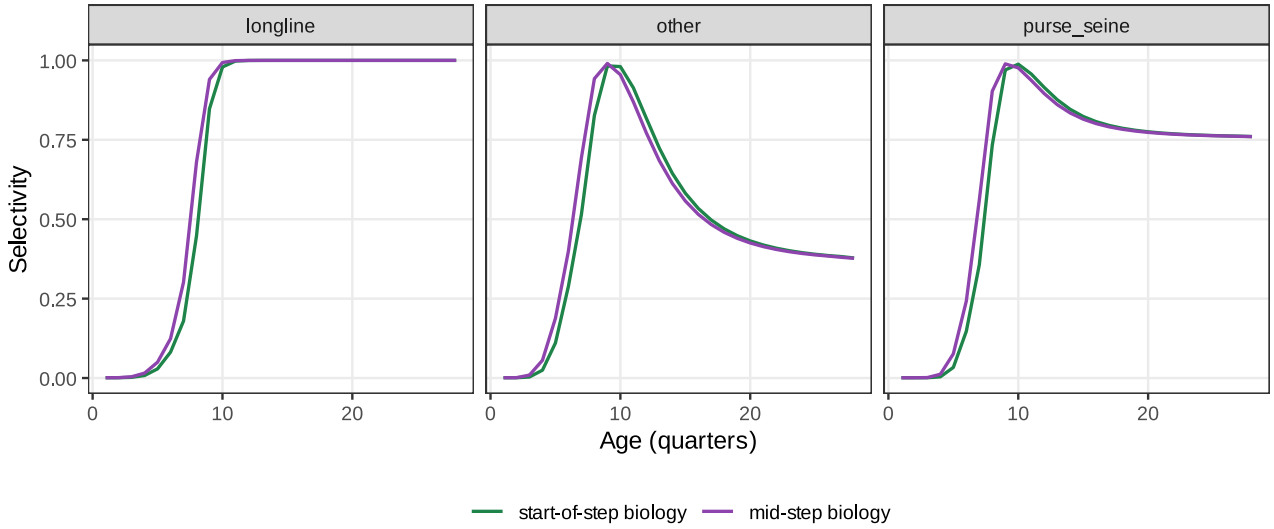


Figure 3: Selectivity-at-age from the same length-based curves, mapped through start-of-quarter or mid-quarter length-at-age.

At the observed catch, removing it at the start of the quarter leaves SSB/B0 20% higher than Baranov does (both with mid-quarter biology) (0.272 against 0.225). A fish caught at the start of the quarter is taken before natural mortality acts on it, so the same catch in numbers costs the stock less. The gap grows with the harvest rate: at 1.25 × the catch, Baranov drives the stock to collapse while start-of-quarter removal still leaves 0.09 of B0.

Pope's mid-step approximation reproduces Baranov to within 2.4% up to the observed catch, without iteration.

Biology timing matters on its own. With the same length-based curves, mid-quarter length-at-age makes every fishery select slightly younger fish (the figure above), and with Baranov catch terminal depletion moves from 0.195 (start) to 0.225 (mid). The two effects run in opposite directions: opal's combination (`start` and `start`, 0.243) is closer to SS3's (`baranov` and `mid`, 0.225) than either change alone would make it.

4 Part 2: fitting with the wrong convention

4.1 Design

Two operating models generate data:

- **SS3-like:** Baranov catch, mid-quarter biology.
- **opal-like:** start-of-quarter catch and biology.

Each of 30 replicates draws new recruitment deviations (N(0, 0.6), shared by both operating models, redrawn if any quarterly harvest fraction would exceed 0.5) and new observation error:

- a longline index, lognormal with CV 0.2, quarters 81–256;
- multinomial length compositions of each fishery's catch (effective sample size 20) in every quarter with catch;
- catch, known exactly.

Six estimation models fit each dataset: the three catch methods crossed with the two biology timings. They estimate log B0, 256 recruitment deviations (with a N(0, σ_R) penalty), log q and ten selectivity parameters. Growth, M, steepness and σ_R are fixed at the truth. Every fit starts at the true values and is polished with Newton steps after `nlminb`.

Two caveats. First, the matched estimation model is not exactly unbiased: the penalised recruitment deviations shrink towards zero where the data are weak, which pushes B0 up and depletion down (the reason SS3 has its bias-adjustment ramp). This affects every estimation model alike, so the comparisons that matter are **paired against the matched model on the same dataset**. Second, three pooled fisheries and known growth make this a cleaner problem than the full IO yellowfin assessment.

fits	failed	max |gradient| (median)	max |gradient| > 1e-3	seconds (median)
360	0	0	19	21.273

4.2 Results

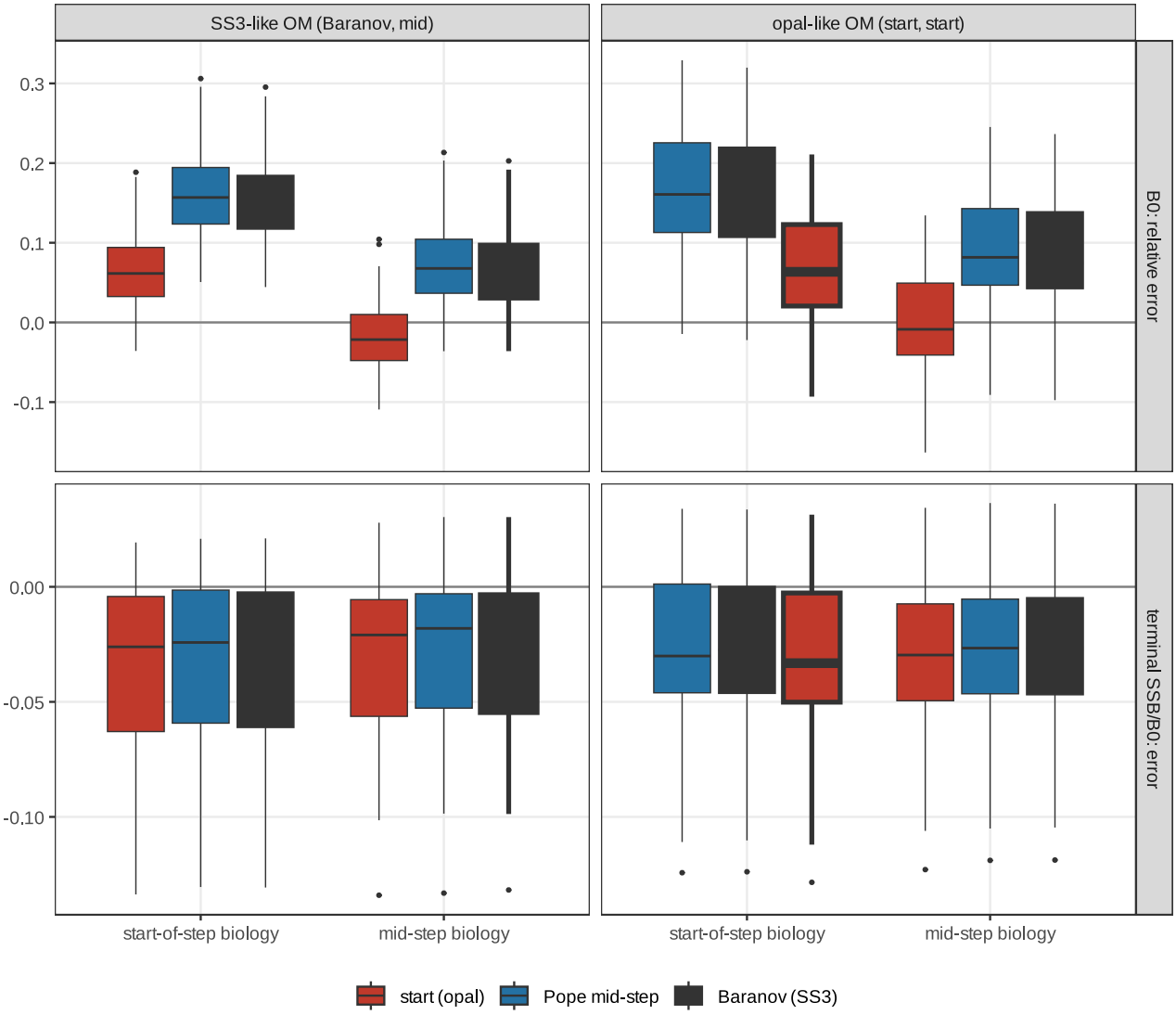


Figure 4: Relative error in B0 (top) and error in terminal depletion (bottom) by estimation model, for each operating model. Boxes span the replicates; the matched model is outlined.

OM	EM catch	EM biology	B0 error	B0 vs matched	Term. dep. error	Term. dep. vs matched	Mean abs. dep. error
SS3-like	baranov *	mid	5.9%	0.0%	−0.019	0.000	0.060
SS3-like	pope_mid	mid	6.8%	0.7%	−0.018	0.000	0.060
SS3-like	start	mid	−2.2%	−7.7%	−0.021	−0.002	0.061
SS3-like	baranov	start	14.8%	8.3%	−0.025	−0.002	0.056
SS3-like	pope_mid	start	15.7%	9.0%	−0.024	−0.002	0.056
SS3-like	start	start	6.1%	0.1%	−0.026	−0.005	0.057

OM	EM catch	EM biology	B0 error	B0 vs matched	Term. dep. error	Term. dep. vs matched	Mean abs. dep. error
opal-like	baranov	mid	7.4%	1.3%	−0.027	0.003	0.064
opal-like	pope_mid	mid	8.2%	2.0%	−0.027	0.003	0.064
opal-like	start	mid	−0.9%	−6.4%	−0.030	0.002	0.065
opal-like	baranov	start	15.2%	8.5%	−0.027	0.003	0.055
opal-like	pope_mid	start	16.1%	9.1%	−0.030	0.003	0.055
opal-like	start *	start	6.4%	0.0%	−0.033	0.000	0.056

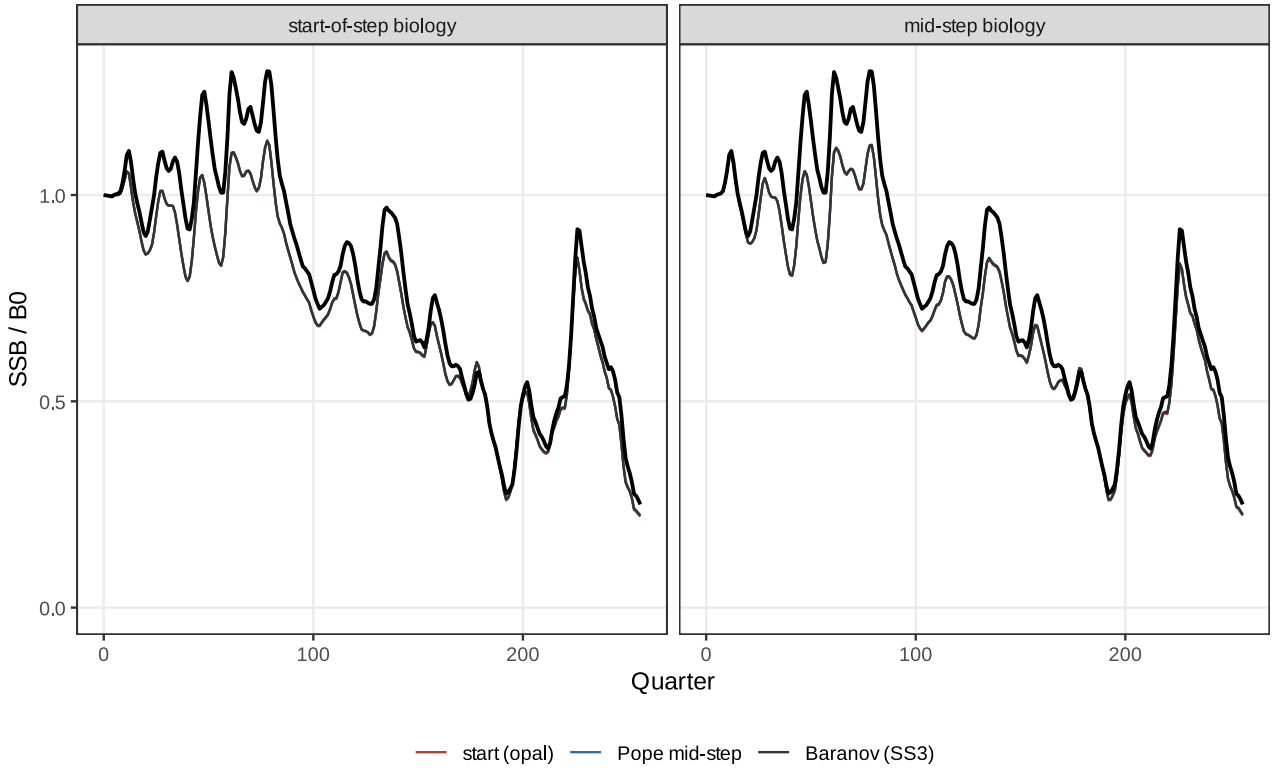


Figure 5: Estimated depletion for replicate 1 against the truth (black), under the SS3-like operating model.

B0 absorbs the convention. Against the matched model on the same dataset, the estimated B0 moves by a consistent amount for each convention, whichever operating model generated the data:

- **Catch:** Baranov catch gives a B0 8.3% to 8.5% higher than start-of-quarter catch with the same biology timing (medians over replicates, for both operating models and both biology timings).
- **Biology:** start-of-quarter biology gives a B0 6.8% to 8.3% higher than mid-quarter biology with the same catch method.
- **Together they cancel.** opal's `start`/`start` fitted to SS3-like data gives a B0 0.1% from the matched model, and `baranov`/`mid` fitted to opal-like data 1.3%. Changing only one of the two conventions moves B0 by 6.8% to 8.5%.
- **Pope tracks Baranov** to within 0.7%.

Estimated depletion barely moves. Every estimation model's terminal depletion is within 0.005 of the matched model's in the median, and within 0.016 in 90% of replicates. Fitted to an index and length compositions, a model with the "wrong" timing rescales B0 so that its relative trajectory matches the data; the convention ends up in the scale, not in the trend. This is the opposite of Part 1, where B0 was held fixed and depletion moved by 20%.

All estimation models share the same small depletion bias against the truth (median −0.025) and B0 bias (6% for the matched models), from recruitment-deviation shrinkage.

5 Implications for opal

1. **Within a quarterly model, these conventions matter for scale, not for depletion.** They move estimated B0 by up to about 8% each, but estimated depletion by less than 0.01. The IO yellowfin difference in depletion between opal and SS3 (0.307 against 0.399 for fit C2) is therefore unlikely to come from catch or biology timing; the C2 vignette traced it to selectivity, and fit A1 (SS3′s selectivity) matched SS3′s depletion.
2. **Where they do matter:** anything evaluated at fixed parameters rather than fitted, such as bridging runs, projections under a fixed catch, catch-based reference points and absolute biomass. Part 1 shows 20–25% in depletion for the same catch, more at higher harvest rates, and longer (annual) time steps would make both effects larger.
3. **Change both or neither.** opal's start-of-quarter catch and start-of-quarter biology roughly cancel against SS3′s Baranov catch and mid-season biology. Adopting only one SS3 convention would move opal's B0 further from SS3′s (about 8%) than it is now.
4. **Pope's mid-step catch is a cheap substitute for Baranov.** It is explicit (no iteration), as cheap as opal's current method, and matched Baranov within about 2% in both parts. If opal adds an SS3-compatible option, Pope mid-step catch together with mid-step length-at-age and mid-step index abundance would reproduce SS3′s timing without solving for F.
5. **Start-of-step removal is the most forgiving at high harvest rates.** For the same catch it leaves the most fish, so near collapse it can show a viable stock where Baranov does not (1.25 × catch in Part 1). This matters most for projections under fixed catches.

6 Reproducing

Step	Script	Does	Time
1	`01-om-inputs.r`	Builds the operating-model inputs from the IO yellowfin export	~30 s
2	`02-deterministic.r`	Part 1 projections	~10 s
3	`03-sim-est.r [n_rep] [n_cores]`	Part 2, 30 replicates × 12 fits	41 min on 4 cores
4	`timing-sim.qmd`	This document; reads only `data/`	seconds

The model is in `R/timing-model.R`. It needs RTMB and opal (for its selectivity curves and `posfun()`); step 1 also reads the stored C2 fit. In a fresh cloud container, `code/rtmb/setup/install-r-opal.sh` installs both. Results were generated 2026-10-14 07:57:30 UTC.